Chemicals & Life

Exam code: 0610 & 0970

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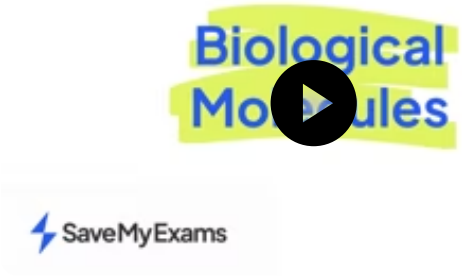
Help me understand



Did this video help you? Yes No

Chemical Elements

- Most of the molecules in living organisms fall into three categories: **carbohydrates, proteins and lipids**
- These **all contain carbon** and so are described as organic molecules

Chemical Elements Table

MOLECULE	CHEMICAL ELEMENTS
CARBOHYDRATE	CARBON, OXYGEN AND HYDROGEN
PROTEIN	ALL CONTAIN CARBON, OXYGEN, HYDROGEN AND NITROGEN AND SOME CONTAIN SMALL AMOUNTS OF OTHER ELEMENTS SUCH AS SULPHUR
LIPID	CARBON, OXYGEN AND HYDROGEN

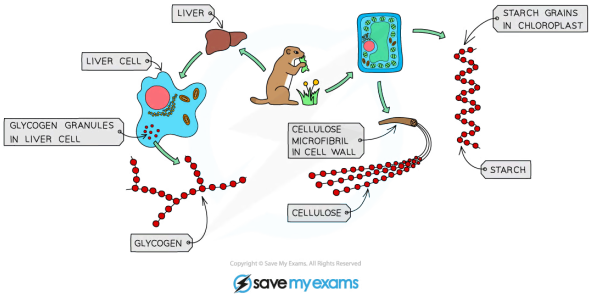
Large Molecules are Made from Smaller Molecules

Carbohydrates

- Long chains of **simple sugars**
- **Glucose** is a simple sugar (a monosaccharide)

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

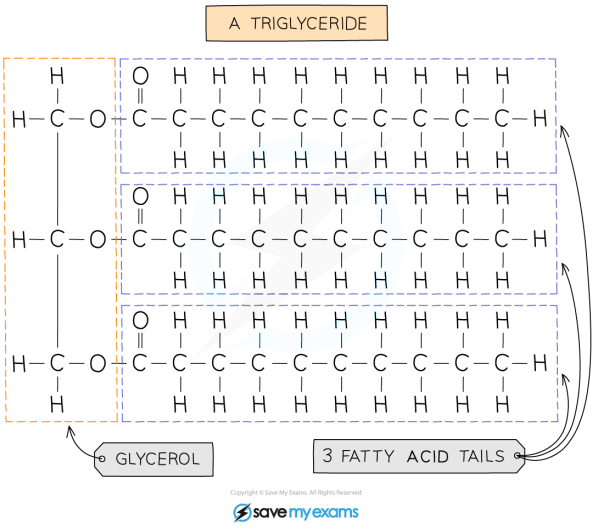
glycogen or cellulose can form (a polysaccharide)



Glycogen, cellulose and starch are all made from glucose molecules

Fats

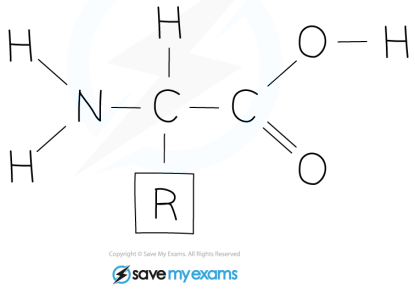
- Most fats (lipids) in the body are made up of **triglycerides**
- Their basic unit is **1 glycerol molecule chemically bonded to 3 fatty acid chains**
- The fatty acids vary in size and structure
- Lipids are divided into **fats** (solids at room temperature) and **oils** (liquids at room temperature)



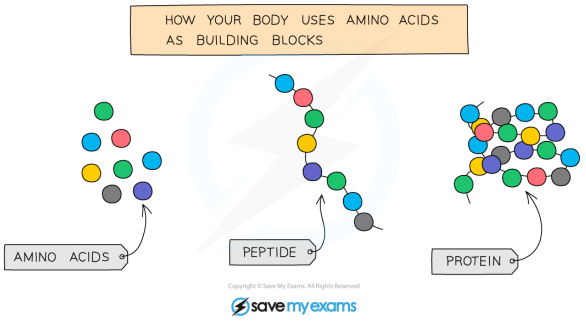
Structure of a triglyceride

Proteins

- Long chains of **amino acids**
- There are about 20 different amino acids
- They all contain the **same basic structure** but the **'R' group is different for each one**
- When amino acids are joined together a protein is formed
- The amino acids can be arranged in any order, resulting in hundreds of thousands of different proteins
- Even a small difference in the order of the amino acids results in a different protein being formed



General amino acid structure



Amino acids join together to form proteins

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